Advanced Causal Inference

Lecture 07: OR, DR Estimators

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This lecture note is partially based on the lecture notes/textbook of Hyunseung Kang, Ying Zhou, Hernán and Robins (2020)

Outline

1. Outcome Regression-based Estimator
2. Doubly Robust Estimator
3. Recent Advances
4. Assignment
5. Estimation of the ATT
6. Sensitivity Analysis

Causal Assumptions in Observational Studies

Assumption (SUTVA (Rubin, 1980))

- $Y = Y(A) = AY(1) + (1 - A)Y(0)$

Assumption (Conditional Ignorability)

- $Y(a) \perp\!\!\!\perp A \,|\, X,\ \forall a$

Assumption (Positivity/Overlap)

- $\Pr(A = a \,|\, X) \in (0, 1),\ \forall (a, X)$

- Under these assumptions,

$$\begin{aligned}
\tau_{\mathsf{ATE}} &= \mathrm{E}\{Y(1) - Y(0)\} \\
&= \mathrm{E}\big\{\mathrm{E}(Y|A = 1, X) - \mathrm{E}(Y|A = 0, X)\big\} && \text{OR-based Representation} \\
&= \mathrm{E}\left\{\frac{AY}{\Pr(A = 1|X)} - \frac{(1 - A)Y}{\Pr(A = 0|X)}\right\} && \text{IPW-based Representation}
\end{aligned}$$

Outcome Regression-based Estimators

- We focus on

$$\tau_{\mathsf{ATE}} = \mathrm{E}\big\{\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X)\big\}$$

- Suppose the following linear model:

$$\mathrm{E}(Y|A,X) = \beta_0 + \beta_a A + \beta_x X$$

- In the regression class, we call β_a as the main effect of A on Y
- Is this the causal effect?
- $\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X) = \beta_a$
- $\tau_{\mathsf{ATE}} = \mathrm{E}\{\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X)\} = \beta_a$
- Is this identification result parametric or nonparametric?

OR-based Estimators

- Suppose the following linear model:

$$\mathrm{E}(Y|A,X) = \beta_0 + \beta_a A + \beta_x X + \beta_{ax} AX$$

- Is this the causal effect?
- $\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X) = \beta_a + \beta_{ax} X$
- $\tau_{\mathsf{ATE}} = \mathrm{E}\{\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X)\} = \beta_a + \beta_{ax}\mathrm{E}(X)$
- If $\overline{X} = 0$, $\tau_a = \beta_a$
- Is this identification result parametric or nonparametric?

OR-based Estimators

- $\tau_{\mathsf{ATE}} = \mathrm{E}\big\{\mathrm{E}(Y|A=1,X) - \mathrm{E}(Y|A=0,X)\big\}$

- An OR-based estimator can be obtained:

1. Estimate $\mu(a,X) = E(Y|A=a,X)$ based on statistical methods
   (e.g., OLS, WLS, logistic regression, local linear regression, machine learning, etc)

2. Based on the plug-in principle:

$$\widehat{\tau}_{\mathsf{OR}} = \frac{1}{N}\sum_{i=1}^{N}\left\{\widehat{\mu}(1,X_i) - \widehat{\mu}(0,X_i)\right\}$$

OR-based Estimators

- How can we conduct hypothesis testing and construct a confidence interval for τ_{ATE}?
  - $\texttt{lm}(Y \sim A + X) \quad \Rightarrow \quad \widehat{\mathrm{Var}}(\hat{\tau}_{\mathsf{OR}}) = \widehat{\mathrm{Var}}(\hat{\beta}_a)$
  - $\texttt{lm}(Y \sim A + X + A:X) \quad \Rightarrow \quad \widehat{\mathrm{Var}}(\hat{\tau}_{\mathsf{OR}}) = \widehat{\mathrm{Var}}(\hat{\beta}_a + \overline{X}^\intercal \hat{\beta}_{ax}) = \widehat{\mathrm{Var}}(\hat{\beta}_a) + 2\overline{X}^\intercal \widehat{\mathrm{Cov}}(\hat{\beta}_{ax}, \hat{\beta}_a) + \overline{X}^\intercal \widehat{\mathrm{Var}}(\hat{\beta}_{ax}) \overline{X}$

- In general, one can use the M-estimation theory, GMM-estimation theory, or bootstrap

- If the outcome model is correctly specified, the OR-based estimator is typically more efficient than the IPW estimator, since it is often the MLE (Scharfstein et al., 1999b; Bang and Robins, 2005)

Example: NHEFS

- `causaldata::nhefs_complete`:

  National Health and Nutrition Examination Survey Epidemiologic Follow-up Study

Role	Variable	Description	Example Values / Notes
X	sex	Sex	0 = Female, 1 = Male
	race	Race	1 = White, 2 = Black, 3 = Other
	age	Age at baseline	e.g., 25–74
	school	Years of schooling	e.g., 0–20
	smokeintensity	Number of cigarettes smoked per day in 1971	e.g., 0–60
	smokeyrs	Years of smoking	e.g., 0–50
	exercise	Recreational exercise in 1971	0 = much, 1 = moderate, 2 = little/no
	active	Usual daily activity in 1971	0 = very active, 1 = moderately, 2 = inactive, 3 = missing
	wt71	Weight in kilograms in 1971	e.g., 45–120
A	qsmk	Quit smoking between 1st questionnaire and 1982	0 = No, 1 = Yes
Y	wt82	Weight in kilograms in 1982	e.g., 45–130
	wt82_71	Weight change (wt82 - wt71)	Can be negative or positive

- Study the effect of quit smoking on weight in 1982

Example: NHEFS

- First, we consider $\mathrm{E}(Y|A,X) = \beta_0 + \beta_a A + \beta_x X$:

```
OLS.1 <- lm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,
            data=data)
c( coef(summary(OLS.1))[2,1:2],
   coef(summary(OLS.1))[2,1]-1.96*coef(summary(OLS.1))[2,2],
   coef(summary(OLS.1))[2,1]+1.96*coef(summary(OLS.1))[2,2])
```

- $\widehat{\tau}_{\mathsf{OR}} = 3.328$, $\mathsf{SE} = 0.441$, 95% $\mathsf{CI} = (2.464, 4.192)$

Example: NHEFS

- Second, we consider $\mathrm{E}(Y|A,X) = \beta_0 + \beta_a A + \beta_x X + \beta_{ax} AX$:

```
FORMULA <- as.formula(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71+
                        qsmk:(sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71))
OLS.2 <- lm(FORMULA, data=data)
data.1 <- data.0 <- data
data.1$qsmk <- 1; data.0$qsmk <- 0
ATE.Est <- mean( predict(OLS.2, newdata=data.1) - predict(OLS.2, newdata=data.0) )
xmean <- apply(model.matrix(FORMULA, data=data.1),2,mean)

vec <- c(0,1,rep(0,11), xmean[14:24])
SD.ATE <- sqrt(t(vec)%*%vcov(OLS.2)%*%(vec))

c(ATE.Est, SD.ATE, ATE.Est - 1.96*SD.ATE, ATE.Est + 1.96*SD.ATE)
```

- $\widehat{\tau}_{\mathsf{OR}} = 3.375$, SE $= 0.457$, 95% CI $= (2.479, 4.270)$

Example: NHEFS

- Third, we consider $\mathrm{E}(Y|A,X) = \mathtt{rf}(A,X)$:

```
library(randomForest)
RF <- randomForest(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data)
data.1 <- data.0 <- data
data.1$qsmk <- 1; data.0$qsmk <- 0
ATE.Est <- mean(predict(RF,data.1) - predict(RF,newdata=data.0))

ATE.Boot.Est <- rep(0,100)
for(bb in 1:100){
  boot.index <- sample(nrow(data),nrow(data),replace=TRUE)
  data.boot <- data[boot.index,]
  RF <- randomForest(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data.boot)

  data.boot.1 <- data.boot.0 <- data.boot
  data.boot.1$qsmk <- 1; data.boot.0$qsmk <- 0
  ATE.Boot.Est[bb] <- mean(predict(RF,data.boot.1) - predict(RF,newdata=data.boot.0))
}

c(ATE.Est, sd(ATE.Boot.Est), quantile(ATE.Boot.Est,c(0.025,0.975)))
```

- $\widehat{\tau}_{\mathsf{OR}} = 2.487$, $\mathsf{SE} = 0.366$, 95% $\mathsf{CI} = (1.646,\ 3.068)$

Example: NHEFS

- Lastly, we consider $\mathrm{E}(Y|A,X) = \texttt{gbm}(A,X)$:

```
library(gbm)
GBM <- gbm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data)
data.1 <- data.0 <- data
data.1$qsmk <- 1; data.0$qsmk <- 0
ATE.Est <- mean(predict(GBM,data.1) - predict(GBM,newdata=data.0))

ATE.Boot.Est <- rep(0,100)
for(bb in 1:100){
  boot.index <- sample(nrow(data),nrow(data),replace=TRUE)
  data.boot <- data[boot.index,]
  GBM <- gbm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data.boot)
  data.boot.1 <- data.boot.0 <- data.boot
  data.boot.1$qsmk <- 1; data.boot.0$qsmk <- 0
  ATE.Boot.Est[bb] <- mean(predict(RF,data.boot.1) - predict(RF,newdata=data.boot.0))
}
c(ATE.Est, sd(ATE.Boot.Est), quantile(ATE.Boot.Est,c(0.025,0.975)))
```

- $\widehat{\tau}_{\mathsf{OR}} = 2.179$, SE $= 0.075$, 95% CI $= (2.335, 2.631)$

Another Example

- Consider the following DGP:

$$X \sim N(0,1)$$

$$A = \mathbb{1}(X > 0)$$

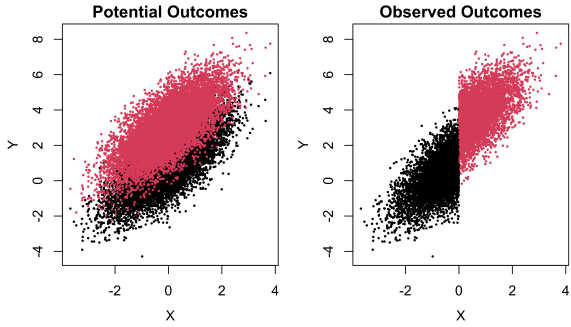
$$Y(1) = \mu(1, X) + \epsilon_1 \ , \quad Y(0) = \mu(0, X) + \epsilon_0 \ , \quad \epsilon_a \perp\!\!\!\perp (A, X) \ , \quad \epsilon_a \sim N(0,1)$$

$$Y = Y(A)$$

  - Are Conditional Ignorability and Positivity satisfied?

Another Example

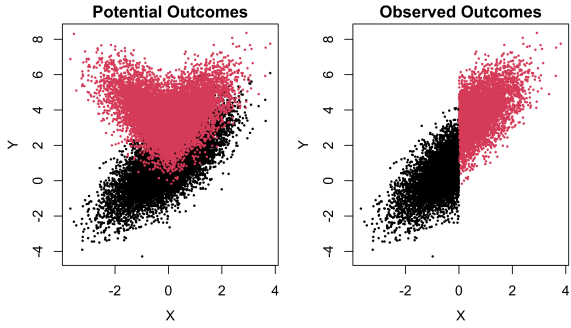
- $\mu(1, X) = 3 + X$ and $\mu(0, X) = 2 + X$



- $\mathrm{E}\{Y(1) - Y(0)\} = \mathrm{E}(2 + \epsilon_1 - \epsilon_0) = 2$
- $\texttt{lm}(Y \sim A + X) \quad \Rightarrow \quad \widehat{\beta}_a = 2.0377$ (SE=0.037) $\simeq 2$

Another Example

- $\mu(1, X) = 3 + |X|$ and $\mu(0, X) = 2 + X$



- $\mathrm{E}\{Y(1) - Y(0)\} = \mathrm{E}(2 + |X| - X + \epsilon_1 - \epsilon_0) = 2.798$
- $\texttt{lm}(Y \sim A + X) \quad \Rightarrow \quad \widehat{\beta}_a = 2.0377$ (SE=0.037) $\neq 2.798$
- $\texttt{lm}(Y \sim A + X + A : X) \quad \Rightarrow \quad \widehat{\tau}_{\mathsf{OR}} = 1.9595$ (SE=0.034) $\neq 2.798$

Another Example

- $\texttt{lm}(Y \sim A + X)$: No interaction term AX in the model; assuming a homogeneous effect
- $\texttt{lm}(Y \sim A + X + A:X)$: Allows for a heterogeneous effect
  - But if there are no observations for $(A = 1, X < 0)$ or $(A = 1, X > 0)$, the interaction cannot be reliably estimated

- An OR-based approach estimates the ATE by extrapolating the specified regression model
  - If the model is correctly specified, the estimator is valid
  - If the model is misspecified, the estimator gives misleading results — this is the cost of lacking overlap

- The OR-based ATE estimator does not use the propensity score

$$f(Y, A, X) = f(Y|A, X)\ f(X)\ f(A|X)$$

$$\begin{cases} f(Y|A, X) & \text{is used to estimate } \mathrm{E}(Y|A, X) \\ f(X) & \text{is used in } N^{-1} \sum_{i=1}^{N} \widehat{\mu}(a, X_i) \end{cases}$$

- It provides no warning about violations of overlap (even though they can be assessed from the observed data)

- The OR-based approach is not a panacea; you need to understand its assumptions and limitations

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Recap: Two identification formulas

- Two identification formulas for $\tau = E\{Y(1) - Y(0)\}$

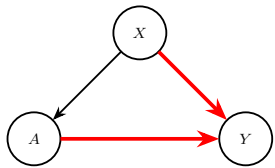
- Outcome Regression

$$\tau = E\{\mu_1(X) - \mu_0(X)\} \tag{1}$$

where $\mu_a(X) = E\{Y(a) \mid X\} = E(Y \mid A = a, X)$

- The outcome regression estimator requires fitting a model for the outcome given the treatment and covariates

- It is consistent if the outcome model is correctly specified



Recap: Two identification formulas

- Inverse Probability Weighting

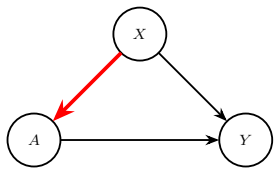
$$\tau = E\left\{\frac{AY}{e(X)} - \frac{(1-A)Y}{1-e(X)}\right\} \tag{2}$$

  where $e(X) = \Pr(A = 1 \mid X)$ is the propensity score

- The IPW estimator requires fitting a model for the treatment given the covariates

- It is consistent if the propensity score model is correctly specified

- There are many combinations of (1) and (2) that lead to different identification formulas of the ATE



Doubly Robust Estimation: Population Version

Define

$$\tilde{\mu}_1^{\mathsf{dr}} = E\left[\frac{A\{Y-\mu_1(X,\beta_1)\}}{e(X,\alpha)} + \mu_1(X,\beta_1)\right], \tag{3}$$

$$\tilde{\mu}_0^{\mathsf{dr}} = E\left[\frac{(1-A)\{Y-\mu_0(X,\beta_0)\}}{1-e(X,\alpha)} + \mu_0(X,\beta_0)\right], \tag{4}$$

which can also be written as

$$\tilde{\mu}_1^{\mathsf{dr}} = E\left[\frac{AY}{e(X,\alpha)} - \frac{A-e(X,\alpha)}{e(X,\alpha)}\mu_1(X,\beta_1)\right], \tag{5}$$

$$\tilde{\mu}_0^{\mathsf{dr}} = E\left[\frac{(1-A)Y}{1-e(X,\alpha)} - \frac{e(X,\alpha)-A}{1-e(X,\alpha)}\mu_0(X,\beta_0)\right]. \tag{6}$$

- (3) and (4): augment the OR estimator by inverse propensity score weighting terms of the residuals
- (5) and (6): augment the IPW estimator by the imputed outcomes
- also called the *augmented inverse propensity score weighting* (AIPW) estimator.

Doubly Robustness Property

- Let's look at $\tilde{\mu}_1^{\mathsf{dr}}$. It has the following property

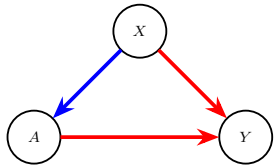
- If either $e(X,\alpha) = e(X)$ or $\mu_1(X,\beta_1) = \mu_1(X)$, then $\tilde{\mu}_1^{\mathsf{dr}} = E\{Y(1)\}$

$$
\begin{aligned}
&\tilde{\mu}_1^{\mathsf{dr}} - E\{Y(1)\} \\
&= E\left[\frac{A\{Y(1)-\mu_1(X,\beta_1)\}}{e(X,\alpha)} - \{Y(1)-\mu_1(X,\beta_1)\}\right] && \text{(by definition)} \\
&= E\left[\frac{A-e(X,\alpha)}{e(X,\alpha)}\{Y(1)-\mu_1(X,\beta_1)\}\right] && \text{(combining terms)} \\
&= E\left(E\left[\frac{A-e(X,\alpha)}{e(X,\alpha)}\{Y(1)-\mu_1(X,\beta_1)\}\,\middle|\, X\right]\right) && \text{(tower property)} \\
&= E\left[E\left\{\frac{A-e(X,\alpha)}{e(X,\alpha)}\,\middle|\, X\right\} \times E\left\{Y(1)-\mu_1(X,\beta_1) \mid X\right\}\right] && \text{(ignorability)} \\
&= E\left[\frac{e(X)-e(X,\alpha)}{e(X,\alpha)} \times \{\mu_1(X)-\mu_1(X,\beta_1)\}\right]
\end{aligned}
$$

- Therefore, $\tilde{\mu}_1^{\mathsf{dr}} - E\{Y(1)\} = 0$ if either $e(X,\alpha) = e(X)$ or $\mu_1(X,\beta_1) = \mu_1(X)$

Doubly Robustness Property

- If either $e(X, \alpha) = e(X)$ or $\mu_a(X, \beta_a) = \mu_a(X)$ for $a = 0, 1$, then $\tilde{\mu}_1^{\text{dr}} - \tilde{\mu}_0^{\text{dr}} = \tau$



AIPW Estimator

- Based on the data $(X_i, A_i, Y_i)_{i=1}^N$, we can obtain a doubly robust estimator for τ by the following steps:

1. obtain the fitted values of the propensity scores: $e(X_i, \widehat{\alpha})$;

2. obtain the fitted values of the outcome means: $\mu_1(X_i, \widehat{\beta}_1)$ and $\mu_0(X_i, \widehat{\beta}_0)$;

3. construct the AIPW estimator: $\widehat{\tau}^{\mathsf{dr}} = \widehat{\mu}_1^{\mathsf{dr}} - \widehat{\mu}_0^{\mathsf{dr}}$, where

$$\widehat{\mu}_1^{\mathsf{dr}} = \frac{1}{n} \sum_{i=1}^{N} \left[\frac{A_i\{Y_i - \mu_1(X_i, \widehat{\beta}_1)\}}{e(X_i, \widehat{\alpha})} + \mu_1(X_i, \widehat{\beta}_1) \right]$$

and

$$\widehat{\mu}_0^{\mathsf{dr}} = \frac{1}{n} \sum_{i=1}^{N} \left[\frac{(1 - A_i)\{Y_i - \mu_0(X_i, \widehat{\beta}_0)\}}{1 - e(X_i, \widehat{\alpha})} + \mu_0(X_i, \widehat{\beta}_0) \right].$$

Inference

- M-estimation theory with the following moment function:

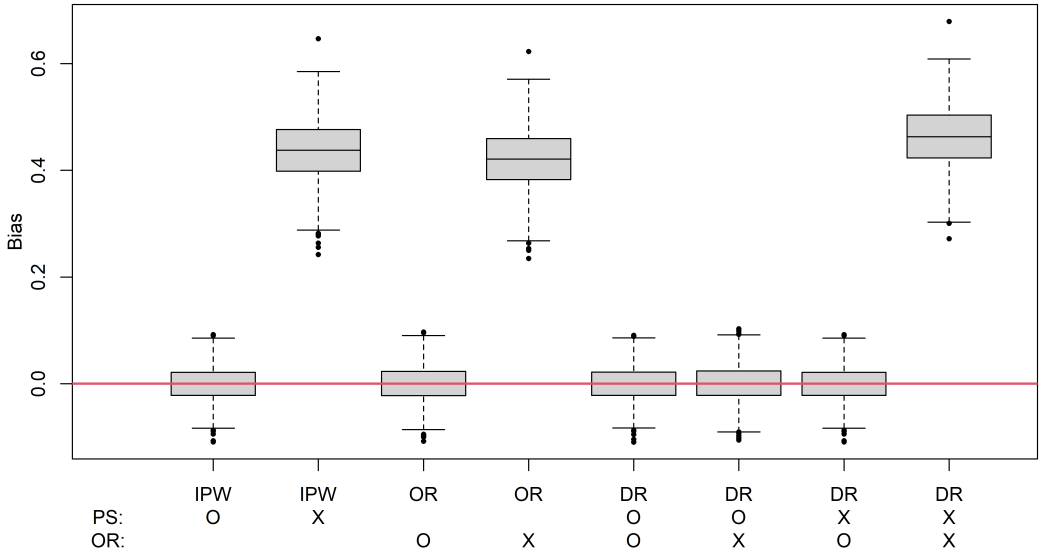
$$\Psi(O;\tau,\alpha,\beta) = \begin{bmatrix} \dfrac{A_i\{Y_i - \mu(1,X_i,\beta)\}}{e(X_i,\alpha)} - \dfrac{(1-A_i)\{Y_i - \mu(0,X_i,\beta)\}}{1-e(X_i,\alpha)} - \tau \\ \nabla_\alpha e(X_i,\alpha)\{A_i - e(X_i,\alpha)\} \\ \nabla_\beta \mu(A_i,\beta)\{Y_i - \mu(A_i,\beta)\} \end{bmatrix}$$

- Or nonparametric bootstrap

AIPW Estimator

- Double Robustness: $\widehat{\tau}^{\text{dr}}$ is a consistent estimator of the ATE if either the propensity score model or the potential outcome model is, but not necessary both are, correctly specified
- DR is a large sample property
- Offers protection against model mis-specification: gives you two chances to get it right (and wrong)!
- If $e(X)$ and $\mu_a(X)$ are modeled correctly, $\widehat{\tau}^{\text{dr}}$ will have a smaller asymp. variance than the IPW estimator
- If the outcome model $\mu_a(X)$ is correct, $\widehat{\tau}^{\text{dr}}$ has a larger asymp. variance than the OR-based estimator
- ... but gives protection in the event that $\mu_a(X)$ is misspecified

AIPW Estimator



Example: NHEFS

```
data <- nhefs_complete
PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")
OLS <- lm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data)
data1 <- data0 <- data;  data1$qsmk <- 1; data0$qsmk <- 0
OR1.Est <- predict(OLS,newdata=data1); OR0.Est <- predict(OLS,newdata=data0)

A <- data$qsmk; Y <- data$wt82

ATE.DR <- mean(A*(Y-OR1.Est)/PS.Est-(1-A)*(Y-OR0.Est)/(1-PS.Est) + (OR1.Est-OR0.Est))
ATE.OR <- mean(OR1.Est-OR0.Est)
ATE.IPW <- mean(A*Y/PS.Est-(1-A)*Y/(1-PS.Est))

ATE <- c(ATE.IPW, ATE.OR, ATE.DR)
```

Example: NHEFS

```
ATE.Boot <- matrix(0,1000,3)
for(bb in 1:1000){
  Index <- sample(nrow(data),nrow(data),replace=TRUE); data.Boot <- data[Index,]
  PS.Boot <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data.Boot, family="binomial")
  PS.Est.Boot <- predict(PS.Boot,type="response")
  OLS.Boot <- lm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data.Boot)
  data1.Boot <- data0.Boot <- data.Boot; data1.Boot$qsmk <- 1; data0.Boot$qsmk <- 0

  OR1.Est.Boot <- predict(OLS.Boot,newdata=data1.Boot); OR0.Est.Boot <- predict(OLS.Boot,newdata=data0.Boot)
  A.Boot <- data.Boot$qsmk; Y.Boot <- data.Boot$wt82

  ATE.DR.Boot <- mean(A.Boot*(Y.Boot-OR1.Est.Boot)/PS.Est.Boot-(1-A.Boot)*(Y.Boot-OR0.Est.Boot)/(1-PS.Est.Boot) +
                        (OR1.Est.Boot-OR0.Est.Boot))
  ATE.OR.Boot <- mean(OR1.Est.Boot-OR0.Est.Boot)
  ATE.IPW.Boot <- mean(A.Boot*Y.Boot/PS.Est.Boot-(1-A.Boot)*Y.Boot/(1-PS.Est.Boot))
  ATE.Boot[bb,] <- c(ATE.IPW.Boot, ATE.OR.Boot, ATE.DR.Boot)
}
round(rbind(ATE, apply(ATE.Boot,2,sd), apply(ATE.Boot,2,function(t){quantile(t,c(0.025,0.975))}) ),3)
```

	IPW	OR	AIPW
Estimate	2.778	3.328	3.313
SE	0.836	0.469	0.489
95% CI	(1.219, 4.379)	(2.426, 4.267)	(2.393, 4.280)

Intuitive Perspectives

- Reducing the variance of the IPW estimator
- Reducing the bias of the outcome regression estimator

Reducing the Variance of the IPW Estimator

- $\mu_1 = E\left\{\frac{AY}{e(X)}\right\}$ completely ignores the outcome model of Y
- Consistent without assuming an outcome model
- However, if covariates predict Y, a residual-based approach often has lower variance
- With a possibly misspecified outcome model $\mu_1(X, \beta_1)$, we can decompose:

$$\mu_1 = E\{Y(1)\} = E\{Y(1) - \mu_1(X, \beta_1)\} + E\{\mu_1(X, \beta_1)\}$$

- Applying the IPW formula to the first term, we obtain:

$$\mu_1 = E\left[\frac{A\{Y - \mu_1(X, \beta_1)\}}{e(X)} + \mu_1(X, \beta_1)\right]$$

- This holds if e is correct, regardless of μ, but μ reduces the variance

Debiasing the Outcome Regression Estimator

- Starting with an Outcome Regression Estimator: $\tilde{\mu}_1 = E\{\mu_1(X, \beta_1)\}$
- May not be the same as μ_1 if the outcome model is misspecified with bias: $E\{\mu_1(X, \beta_1) - Y(1)\}$
- This bias can be estimated using an IPW estimator if $e(X)$ is correct:

$$B = E\left[\frac{A\{\mu_1(X, \beta_1) - Y\}}{e(X)}\right]$$

- Debiased formula:

$$\mu_1 = \tilde{\mu}_1 - B = E\left[\frac{A\{Y - \mu_1(X, \beta_1)\}}{e(X)} + \mu_1(X, \beta_1)\right]$$

A Historical Note: Connection to Sample Survey

- The generalized regression estimator in survey sampling (Cassel et al., 1976)

- If we assume $Y_i \perp\!\!\!\perp A_i \mid X_i$ (A_i = response to survey), then the bias-corrected estimate is

$$\widehat{\mu}_{BC-OLS} = \frac{1}{N}\sum_{i=1}^{N}\widehat{Y}_i + \frac{1}{N}\sum_{i=1}^{N}\frac{A_i}{\widehat{e}_i}\widehat{\epsilon}_i = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{A_i(Y_i - \widehat{Y}_i)}{\widehat{e}_i} + \widehat{Y}_i\right]$$

  where $\epsilon_i = Y_i - \widehat{Y}_i$

- **Idea:** since we know the true value for respondents ($A_i = 1$), we can use that information to correct the bias we may have, providing that the bias is the same in respondents and non-respondents (given covariate information)

  - Red assumption = Ignorability

Outline

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2. Doubly Robust Estimator
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Kang and Schafer (2007)'s Critique

- Kang and Schafer (2007) considered a simulation example, and shows that when both models are misspecified, the DR estimators may not outperform the OLS estimator
  - (X_1, X_2, X_3, X_4) are i.i.d. $N(0, I)$
  - $Y = 210 + 27.4X_1 + 13.7X_2 + 13.7X_3 + 13.7X_4 + \epsilon$, where $\epsilon \sim N(0, 1)$
  - $e(X) = expit(-X_1 + 0.5X_2 - 0.25X_3 - 0.1X_4)$
  - Instead of observing X we observe
    $W_1 = exp(X_1/2), W_2 = X_2/(1 + exp(X_1)) + 10, W_3 = (X_1X_3/25 + 0.6)^3, W_4 = (X_2 + X_4 + 20)^2$

Kang and Schafer (2007)'s Critique

TABLE 3

Performance of ordinary least-squares regression estimators over 1000 *samples from the artificial population*

Sample size	*y*-model	Method	Bias	% Bias	RMSE	MAE
(a) $n = 200$	Correct	*OLS*	−0.08	−3.4	2.48	1.68
	Incorrect	*OLS*	−0.57	−17.7	3.26	2.24
(b) $n = 1000$	Correct	*OLS*	−0.00	−0.1	1.17	0.79
	Incorrect	*OLS*	−0.84	−56.0	1.72	1.15

TABLE 5

Performance of bias-corrected regression estimators over 1000 *samples from the artificial population*

Sample size	π-model	y-model	Method	Bias	% Bias	RMSE	MAE
(a) $n = 200$	Correct	Correct	*BC-OLS*	−0.08	−3.4	2.48	1.68
		Incorrect	*BC-OLS*	0.25	7.5	3.28	2.17
	Incorrect	Correct	*BC-OLS*	−0.08	−3.3	2.48	1.70
		Incorrect	*BC-OLS*	−5.12	−43.0	12.96	3.54
(b) $n = 1000$	Correct	Correct	*BC-OLS*	0.00	−0.1	1.17	0.79
		Incorrect	*BC-OLS*	0.06	3.4	1.75	1.02
	Incorrect	Correct	*BC-OLS*	−0.02	−1.4	1.49	0.80
		Incorrect	*BC-OLS*	−21.03	−13.5	157.21	5.32

Kang and Schafer (2007)'s Conclusion

Double robustness is an interesting theoretical property that can arise in many ways. It does not necessarily translate into good performance when neither model is correctly specified.

Robins et al. (2007)'s Response

1. Cautionary note

- Robins and Wang (2000): When the variance of the inverse probability weights is very large, the sampling distribution of (their doubly robust) estimator can be markedly skew and highly variable, and a parametric imputation estimator may be preferred

- In Kang and Schafer (2007)'s simulation settings, a small subset of the sample will have extremely large weights relative to the remainder of sample

2. Relationship between OLS and DR estimator

- If we posit a linear inverse probability model

$$e(X) = \frac{1}{\alpha^\intercal X},$$

then the OLS is indeed a DR estimator with respect to this model and the outcome model $\mu_1(X) = \beta^\intercal X$

- The fact that OLS outperforms other DR estimators does not mean DR estimators are all garbage: OLS is itself a DR estimator!

Robins et al. (2007)'s Response

- **Sketch of proof**
- The estimating equation for estimating β in $\mu_1(X) = X^T\beta$ is

$$E\{AX(Y - X^\intercal\beta)\} = 0.$$

- It follows that

$$\begin{aligned}
&\frac{1}{N}\sum_{i=1}^{N} A_i X_i (Y_i - X_i^\intercal \widehat{\beta}) = 0 \\
\Rightarrow\ &\frac{1}{N}\sum_{i=1}^{N} A_i \alpha^\intercal X_i (Y_i - \widehat{\beta}) = 0, \quad \forall \alpha \\
\Rightarrow &\frac{1}{N}\sum_{i=1}^{N} \frac{A_i}{e(X_i)} (Y_i - X_i^\intercal \widehat{\beta}) = 0, \quad \forall e(X) = \frac{1}{\alpha^\intercal X_i}
\end{aligned}$$

- Thus, we have

$$\text{DR estimator} = \frac{1}{N}\sum_{i=1}^{N}\left\{X_i^\intercal\widehat{\beta} + \frac{A}{\widehat{e}(X_i)}(Y_i - X_i^\intercal\widehat{\beta})\right\} = \frac{1}{N}\sum_{i=1}^{N} X_i^\intercal\widehat{\beta} = \frac{1}{N}\sum_{i=1}^{N}\hat{\mu}_1(X_i) = \text{OLS estimator}$$

3. Bounded estimators

- Improve DR estimators so that it has the boundedness property

- Boundedness: the estimator always falls in the parameter space
  e.g., if Y is binary, then the estimator should fall in $[0, 1]$

- Example:
  - The HT estimator $\mathbb{P}_N \frac{AY}{\hat{e}(X)}$ is not bounded
  - The Hajek estimator $\mathbb{P}_N \frac{AY/\hat{e}(X)}{A/\hat{e}(X)}$ is bounded
  - The regression estimator $\mathbb{P}_N \hat{\mu}_1(X)$ is bounded if $\hat{\mu}_1(X)$ is contained in the sample space of Y

Robins et al. (2007)'s Response

- Q: How to construct bounded estimators?
- A: Estimate nuisance parameters in such a way that the final DR estimators is the Hajek estimator or the regression estimator!

Regression DR Estimator & TMLE

- **Goal**: estimate nuisance parameters so that

$$\mathbb{P}_N \left[\widehat{\mu}_1(X) + \frac{A}{\widehat{e}(X)} \{Y - \widehat{\mu}_1(X)\} \right] = \mathbb{P}_N \widehat{\mu}_1(X) \quad \Leftrightarrow \quad \mathbb{P}_N \frac{A}{\widehat{e}(X)} \{Y - \widehat{\mu}_1(X)\} = 0 \tag{7}$$

- Note that if $\widehat{\mu}_1(X) = X^\intercal \widehat{\beta}$, the estimating equation for $\widehat{\mu}_1(X)$ takes the form

$$\mathbb{P}_N A \{Y - \widehat{\mu}_1(X)\} X = 0$$

- Scharfstein et al. (1999a): Include $1/\widehat{e}(X)$ in the regression model, so that it becomes $\mu_1(X) = X^\intercal \beta + \phi \frac{1}{\widehat{e}(X)}$

- Then we solve

$$\mathbb{P}_N A \{Y - \widehat{\mu}_1(X)\} \begin{pmatrix} 1/\widehat{e}(X) \\ X \end{pmatrix} = 0$$

  - The first equation is exactly (7)!
  - Refined into Targeted Maximum Likelihood (aka Minimum Loss-based) Estimation (TMLE; van der Laan and Rubin (2006))

Example: NHEFS

```
library(tmle)
data <- nhefs_complete

PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")

OLS <- lm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,data=data)
data1 <- data0 <- data; data1$qsmk <- 1; data0$qsmk <- 0
OR1.Est <- predict(OLS,newdata=data1); OR0.Est <- predict(OLS,newdata=data0)

tmle(Y=data$wt82,                # outcome
     A=data$qsmk,                # treatment
     W=data[,COV],               # pre-trt covariates
     Q=cbind(OR1.Est,OR0.Est),   # matrix of mu_1, mu_0 estimates
     g1W=PS.Est)                 # propensity score estiamtes
```

- $\widehat{\tau}_{\mathsf{TMLE}} = 3.329$, $\mathsf{SE} = 0.246$, 95% $\mathsf{CI} = (2.357, 4.301)$
- $\widehat{\tau}_{\mathsf{AIPW}} = 3.313$, $\mathsf{SE} = 0.489$, 95% $\mathsf{CI} = (2.393, 4.280)$

Weighted Regression DR Estimator

- Fit weighted least squares for $\mathbb{E}(Y \mid A = 1, X) = \mu_1(X) = X^\intercal \beta$ with inverse probability weights $1/\widehat{e}(X)$

```
lm (Y ~ X, data = Data[A==1,], weights = 1/PS)
lm (Y ~ X, data = Data[A==0,], weights = 1/(1-PS))
```

- $\widehat{\beta}$ solves the estimating equation, and X includes an intercept term:

$$0 = \mathbb{P}_n \frac{A}{\widehat{e}(X)} \{Y - X^\intercal \widehat{\beta}\} X \quad \Rightarrow \quad 0 = \mathbb{P}_n \frac{A}{\widehat{e}(X)} \{Y - X^\intercal \widehat{\beta}\} = 0$$

- Weighted regression estimator $\mathbb{P}_n \widehat{\mu}_1(X)$ is also DR (and bounded)

Example: NHEFS

```
library(causaldata)

data <- nhefs_complete

PS <- glm(qsmk~sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71, data=data, family="binomial")
PS.Est <- predict(PS,type="response")
IPW.Weight <- data$qsmk/PS.Est + (1-data$qsmk)/(1-PS.Est)

WLS <- lm(wt82~qsmk+sex+race+age+school+smokeintensity+smokeyrs+exercise+active+wt71,
          data=data,
          weights=IPW.Weight)
WLS.DR.Est <- coef(summary(WLS))[2,1]
WLS.DR.SE <- coef(summary(WLS))[2,2]
round(c(WLS.DR.Est, WLS.DR.SE, WLS.DR.Est-1.96*WLS.DR.SE, WLS.DR.Est+1.96*WLS.DR.SE),3)
```

- $\widehat{\tau}_{\text{WDR}} = 3.299$, SE $= 0.390$, 95% CI $= (2.535, 4.064)$

Bounded HT Doubly-Robust estimator

- Goal: estimate nuisance parameters so that

$$\mathbb{P}_n \widehat{\mu}_1(X) + \frac{\mathbb{P}_n A\{Y - \widehat{\mu}_1(X)\}/\widehat{e}(X)}{\mathbb{P}_n A/\widehat{e}(X)} = \frac{\mathbb{P}_n AY/\widehat{e}(X)}{\mathbb{P}_n A/\widehat{e}(X)}$$

  - This appears to be related to the Hajek estimator, but Robins et al. (2007) called Horvitz-Thompson

- In other words:

$$\mathbb{P}_n \left[\left\{ \widehat{\mu}_1(X) - \mathbb{P}_n \widehat{\mu}_1(X) \right\} \left\{ \frac{A}{\widehat{e}(X)} - 1 \right\} \right] = 0$$

- Robins et al. (2007): The OLS estimator is also a bounded HT doubly-robust estimator with a linear inverse probability model!

DR as a Diagnostic Tool

- **DR as a diagnostic tool** (Robins et al., 2001): compare DR with IPW and OR estimates of the same data
- If DR is close to IPW but not OR, then the outcome model is likely wrong
- If DR is close to OR but not IPW, then the PS model is likely wrong
- If DR is far away from both IPW and OR, trouble!
  Change models, try more flexible ones

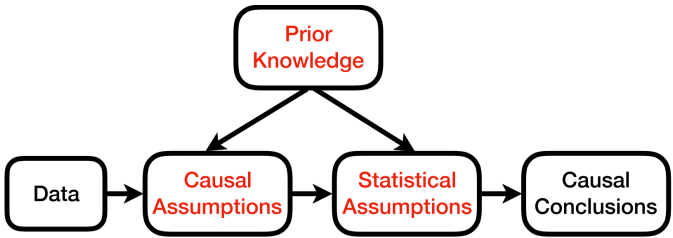
Relationship between the DR Estimator and Efficiency Bound

Relationship between the DR Estimator and Efficiency Bound

- **Covering this topic properly takes more than a few slides**

Summary

- Under the ignorability and positivity assumptions, we analyze observational studies as if it is a conditional randomized experiment
- The propensity score $\Pr(A = 1 \mid X)$ is often known in a cond. rand. exp. but not in an obs. study
- When X is continuous/high-dimensional, we need to model $E(Y \mid A = a, X)$ and/or $P(A = 1 \mid X)$



Outline

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Assignment

- Analyze the dataset `causalweight::JC` (Job Corps)
  While the dataset contains additional variables, we will focus only on the following:
  - $X =$ (`female, age, white, black, hispanic`)
  - $A =$ `assignment`
  - $Y =$ `earny4`

  We aim to study the effect of Job Corps program assignment on weekly earnings in the fourth year after assignment

- Using the dataset, obtain the ATE estimates based on the following:
  - the outcome regression-based estimator;
  - the AIPW estimator;
  - the TMLE;
  - the weighted Regression DR estimator;

- For the AIPW estimator, compute standard errors using both M-estimation theory and the bootstrap method

- Be sure to interpret the point estimate and inference results, as understanding how to draw the causal conclusion is the key focus of this course

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Review: ATT

Assumption (SUTVA (Rubin, 1980))

- $Y = Y(A) = AY(1) + (1 - A)Y(0)$

Assumption (Conditional Ignorability)

- $Y(0) \perp\!\!\!\perp A \,|\, X$

Assumption (Positivity/Overlap)

- $\Pr(A = 1 \,|\, X) < 1, \; \forall X$

IPW/OR/DR Representations of the ATT

- Let

$$\mu(0, X) = \mathrm{E}(Y|A = 0, X)$$
$$w(X) = \frac{\Pr(A = 1|X)}{\Pr(A = 0|X)}$$

- Under these assumptions,

$$\tau_{\mathsf{ATT}} = \mathrm{E}\{Y(1) - Y(0) \,|\, A = 1\} = \frac{1}{\Pr(A = 1)}\mathrm{E}\big\{AY - (1 - A)w(X)Y\big\} \quad \text{(IPW)}$$
$$= \frac{1}{\Pr(A = 1)}\mathrm{E}\big[A\{Y - \mu(0, X)\}\big] \quad \text{(OR)}$$
$$= \frac{1}{\Pr(A = 1)}\mathrm{E}\big[\{A - (1 - A)w(X)\}\{Y - \mu(0, X)\}\big] \quad \text{(DR)}$$

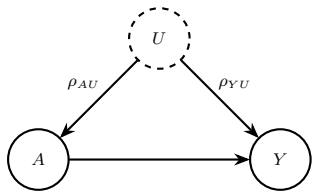
- IPW/OR/DR estimators of τ_{ATT} can be constructed from the plug-in principle

Outline

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E-value

- Omit X for simplicity
- $\rho = \frac{\Pr\{Y(1)=1\}}{\Pr\{Y(0)=1\}}$: (Causal) risk ratio
- $\rho^{\text{obs}} = \frac{\Pr(Y=1|A=1)}{\Pr(Y=1|A=0)}$: (Observed) risk ratio
  - If there is no U, $\rho = \rho^{\text{obs}}$
- ρ_{AU} = risk ratio of U on A; ρ_{YU} = risk ratio of U on Y
- Suppose we estimate $\widehat{\rho}^{\text{obs}} \geqslant 1$ from the IPW, OR, DR estimators
- We wish to assess what happens to $\widehat{\rho}^{\text{obs}}$ when U is present



E-value

- Result 1 (Ding and VanderWeele, 2016):

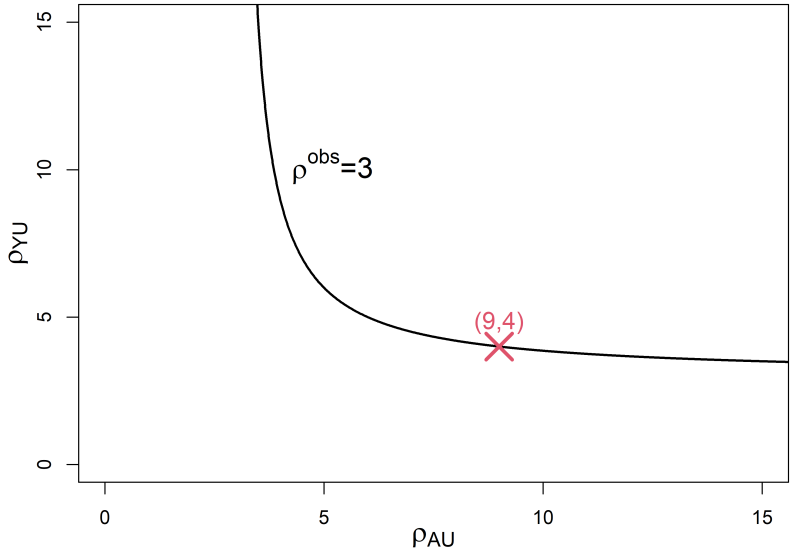
$$\rho \geqslant \frac{\rho^{\text{obs}}}{B} \quad \text{where} \quad B = \frac{\rho_{AU}\rho_{YU}}{\rho_{AU} + \rho_{YU} - 1}$$

  - The bound is sharp and U can be any type

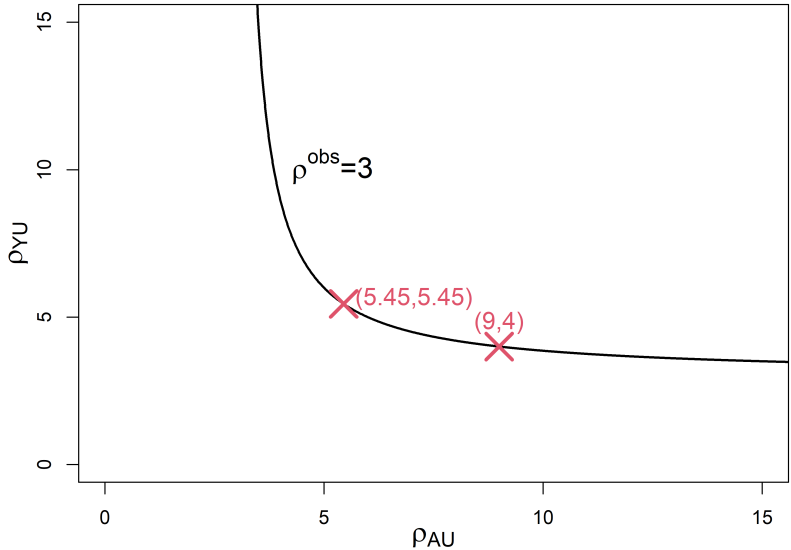
- One can specify values for ρ_{AU} and ρ_{YU}, calculate B, and then check whether $\widehat{\rho}^{\text{obs}}/B$ (or its CI) includes 1

- If large values of ρ_{AU} or ρ_{YU} are required (i.e., a large B) to nullify the observed effect $\widehat{\rho}^{\text{obs}}$, this suggests the finding is relatively robust to the existence of U

E-value



E-value



E-value

- Suppose ρ_{AU} and ρ_{YU} are the same as $\rho_U = \rho_{AU} = \rho_{YU}$

$$B = \frac{\rho_{AU}\rho_{YU}}{\rho_{AU} + \rho_{YU} - 1} = \frac{\rho_U^2}{2\rho_U - 1}$$

- One can specify a value for ρ_U, calculate B, and then check whether $\widehat{\rho}^{\text{obs}}/B$ (or its CI) includes 1

- E-value (VanderWeele and Ding, 2017):

$$\text{E-value} = \widehat{\rho}^{\text{obs}} + \sqrt{\widehat{\rho}^{\text{obs}}(\widehat{\rho}^{\text{obs}} - 1)}$$

E-value

- Cornfield et al. (1959): $\widehat{\rho}^{\text{obs}} = 9.35$ between smoking (A) and the lung cancer death (Y)
- E-value:

$$\text{E-value} = \widehat{\rho}^{\text{obs}} + \sqrt{\widehat{\rho}^{\text{obs}}(\widehat{\rho}^{\text{obs}} - 1)} = 18.18586$$

- To remove the observed effect, U would need to increase the risk of both smoking and lung cancer by at least 18 times
- This is extremely unlikely, so the causal conclusion is pretty robust

E-value

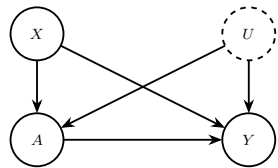
- Pros
  - Simple calculation and easy interpretation
  - No assumptions needed
- Cons
  - Almost linear with effect estimates: E-value $= \hat{\rho}^{\text{obs}} + \sqrt{\hat{\rho}^{\text{obs}}(\hat{\rho}^{\text{obs}} - 1)} \simeq 2\hat{\rho}^{\text{obs}}$
  - Requires context to interpret
  - Low E-value doesn't always mean confounding fully explains association
  - High E-value doesn't always rule out confounding (e.g., confounding by indication)
  - Only addresses confounding that inflates associations

Loss Minimization Approach

- Yadlowsky et al. (2022)

- Assume the latent ignorability:

$$Y(1), Y(0) \perp\!\!\!\perp A \,|\, (X, U)$$



- Following the Rosenbaum (2002)'s sensitivity framework:

$$\frac{1}{\Gamma} \leqslant \frac{\Pr(A=1|X,U=u)}{\Pr(A=0|X,U=u)} \frac{\Pr(A=0|X,U=0)}{\Pr(A=1|X,U=0)} \leqslant \Gamma \quad \Leftrightarrow \quad \text{logit}\{\Pr(A=1|X,U)\} = \alpha(X) + (\log\Gamma)\beta(U,X)$$

- If $\beta(U,X) = \beta(X)$, then $\Pr(A=1|X,U) = \Pr(A=1|X)$, meaning that U is not a confounder

Loss Minimization Approach

- Main assumptions:

$$Y(1), Y(0) \perp\!\!\!\perp A \,\big|\, (X, U) \qquad \text{(Latent Ign.)}$$

$$\frac{1}{\Gamma} \leqslant \frac{\Pr(A=1|X, U=u)}{\Pr(A=0|X, U=u)} \frac{\Pr(A=0|X, U=0)}{\Pr(A=1|X, U=0)} \leqslant \Gamma \qquad (\Gamma\text{-Selection})$$

- Let $\mathcal{Q}_x$ be

$$\mathcal{Q}_x = \left\{ Q = Q\{Y(1), Y(0), A|X=x\} \;\middle|\; \begin{array}{l} Q \text{ satisfies (Latent Ign.) and } (\Gamma\text{-Selection}) \\ Q(A|X) = P^{\text{obs}}(A|X) \text{ and } Q\{Y(1)|A=1, X\} = P^{\text{obs}}(Y|A=1, X) \end{array} \right\}$$

- Let $\theta_1^{\pm}(x)$ be

$$\theta_1^-(x) = \inf_{Q \in \mathcal{Q}_x} \mathrm{E}_Q\{Y(1)|A=0, X=x\} \;, \quad \theta_1^+(x) = \sup_{Q \in \mathcal{Q}_x} \mathrm{E}_Q\{Y(1)|A=0, X=x\}$$

- Let $\theta_0^{\pm}(x)$ be

$$\theta_0^-(x) = \inf_{Q \in \mathcal{Q}_x} \mathrm{E}_Q\{Y(0)|A=1, X=x\} \;, \quad \theta_0^+(x) = \sup_{Q \in \mathcal{Q}_x} \mathrm{E}_Q\{Y(0)|A=1, X=x\}$$

Loss Minimization Approach

- Note that $\theta_1^-(x) = \inf_{Q \in \mathcal{Q}_x} \mathrm{E}_Q\{Y(1)|A=0, X=x\}$ can be expressed as

$$\theta_1^-(\cdot) = \min_\theta \mathrm{E}\Big[\{Y - \theta(X)\}_+^2 + \Gamma\{Y - \theta(X)\}_-^2 \,\Big|\, A = 1\Big]$$

$$\begin{cases} (a)_+ = \max(a, 0) \\ (a)_- = -\min(a, 0) \end{cases}$$

  - Other functions $\theta_1^+(x)$ and $\theta_0^\pm(x)$ can be similarly expressed

- Then, we have a bound of each counterfactual mean:

$$\mathrm{E}\{Y(a)\} \in [\mu_a^-, \mu_a^+]$$

$$\begin{cases} \mu_1^\pm = \mathrm{E}\{AY + (1-A)\theta_1^\pm(X)\} \\ \mu_0^\pm = \mathrm{E}\{(1-A)Y + A\theta_0^\pm(X)\} \end{cases}$$

- Therefore, given Γ, we have a bound on the ATE:

$$\tau^- := \mu_1^- - \mu_0^+ \leqslant \tau \leqslant \tau^+ := \mu_1^+ - \mu_0^-$$

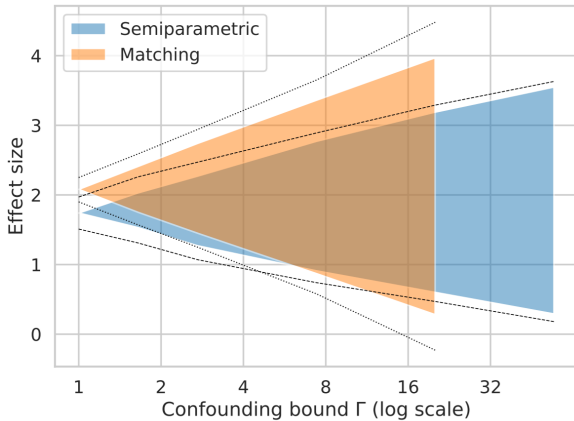
- One can also construct a 95% confidence interval for the ATE (uncertainty interval for the bound) given Γ

$$\widehat{\mathsf{CI}} = \left[\hat{\tau}^- - 1.96\frac{\hat{\sigma}_\tau^-}{\sqrt{N}}, \hat{\tau}^+ + 1.96\frac{\hat{\sigma}_\tau^+}{\sqrt{N}}\right]$$

- To remove the observed effect, Γ is at least 32

Loss Minimization Approach

- The effect of fish consumption on blood mercury levels
- Figure 2 of Yadlowsky et al. (2022)



Omitted Variable Bias Approach

- Chernozhukov et al. (2024)

- Suppose

$$Y(a) = g_Y(a, X, U, \epsilon_Y)$$
$$A = g_A(X, U, \epsilon_A)$$
$$U = g_U(U, \epsilon_U)$$
$$X = \epsilon_X$$

  and latent ign.: $Y(a) \perp\!\!\!\perp A | (X, U)$

- We define "long" and "short" parameters

$$\tau = \mathrm{E}\{\mu(1, X, U) - \mu(0, X, U)\} \qquad \mu(a, X, U) = \mathrm{E}\{Y(a) | A = a, X, U\}$$
$$\tau_s = \mathrm{E}\{\mu_s(1, X) - \mu_s(0, X)\} \qquad \mu_s(a, X) = \mathrm{E}\{Y(a) | A = a, X\}$$

- Omitted variable bias (OVB) $= \tau_s - \tau$
  - OVB $= 0$ if $U = \varnothing$

Omitted Variable Bias Approach

- We also define "long" and "short" IPW terms:

$$w(A,X,U) = \frac{A}{\Pr(A=1|X,U)} - \frac{1-A}{\Pr(A=0|X,U)} \ , \qquad w_s(A,X) = \frac{A}{\Pr(A=1|X)} - \frac{1-A}{\Pr(A=0|X)}$$

- We have

$$\mu_s(A,X) = \mathrm{E}\{\mu(A,X,U)|A,X\} \ , \qquad w_s(A,X) = \mathrm{E}\{w(A,X,U)|A,X\}$$

- Then the bias is characterized as

$$\tau_s - \tau = \mathrm{E}\Big[\{\mu_s(A,X) - \mu(A,X,U)\}\{w_s(A,X) - w(A,X,U)\}\Big]$$

- The sharp bound of the OVB:

$$(\tau_s - \tau)^2 \leqslant \rho^2 B^2 \leqslant B^2 \qquad \begin{cases} B^2 = \mathrm{E}\{(\mu_s - \mu)^2\}\mathrm{E}\{(w_s - w)^2\} \\ \rho^2 = \mathrm{Cov}^2(\mu_s - \mu, w_s - w) \end{cases}$$

Omitted Variable Bias Approach

- The OVB bound: $B^2 = C_Y^2 C_A^2 S^2$ where

$$C_Y^2 = \frac{\mathrm{E}\{(\mu - \mu_s)^2\}}{\mathrm{E}\{(Y - \mu_s)^2\}} \qquad C_A^2 = \frac{\mathrm{E}(w^2) - \mathrm{E}(w_s^2)}{\mathrm{E}(w_s^2)} \qquad S^2 = \mathrm{E}\{(Y - \mu_s)^2\}\mathrm{E}(w_s^2)$$

- Red terms are not identifiable in the observed data, which we can specified as sensitivity parameters

- Interpretation:

$$\begin{aligned} C_Y^2 &= R^2_{Y-\mu_s \sim \mu - \mu_s} \\ &= \text{proportion of residual variance } \mathrm{Var}(Y - \mu_s) \text{ explained by } U \\ &= \text{nonparametric partial } R^2 \text{ of } Y \text{ and } U \text{ given } (A, X) \end{aligned}$$

$$\begin{aligned} \frac{C_A^2}{1 + C_A^2} &= 1 - R^2_{w \sim w_s} \\ &= \frac{\mathrm{E}\{1/\mathrm{Var}(A|X,U)\} - \mathrm{E}\{1/\mathrm{Var}(A|X)\}}{\mathrm{E}\{1/\mathrm{Var}(A|X)\}} \in [0,1] \\ &= \text{relative gain in the average precision of the treatment model due to } U \end{aligned}$$

Omitted Variable Bias Approach

- Therefore, given C_Y and C_A
  - We have a bound of the ATE:

$$\tau \in [\tau^-, \tau^+] \qquad \tau^{\pm} = \tau_s \pm C_Y C_A S$$

  - The estimated bound of the ATE is

$$\tau \in [\hat{\tau}^-, \hat{\tau}^+] \qquad \hat{\tau}^{\pm} = \hat{\tau}_s \pm C_Y C_A \hat{S}$$

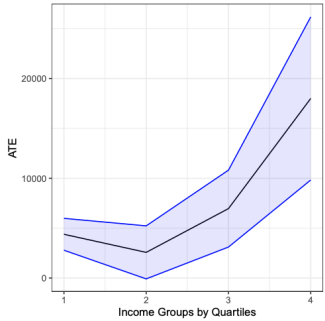
  - One can also construct a 95% confidence interval for the ATE (uncertainty interval for the bound):

$$\widehat{\mathsf{CI}} = \left[\hat{\tau}^- - 1.96\frac{\hat{\sigma}_\tau^-}{\sqrt{N}}, \hat{\tau}^+ + 1.96\frac{\hat{\sigma}_\tau^+}{\sqrt{N}}\right]$$

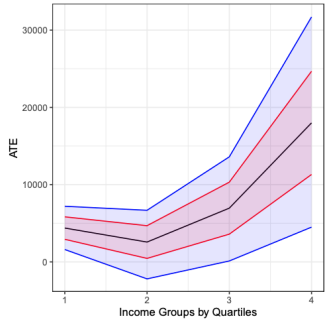
- One can find a value R^* such that $C_Y^* \leqslant R^*$ and $\frac{C_A^2}{1+C_A^2} \leqslant R^*$ and $\widehat{\mathsf{CI}}$ starts to contain zero

Omitted Variable Bias Approach

- The effect of 401(k) eligibility on net financial assets
- We assume that U explain as much of the variation in Y as they do in the total employer 401(k) contributions, which can be up to 5% of an employee's salary
- Figure 3 of Chernozhukov et al. (2024)
- Bounds are still largely positive
- Relatively robust to plausible violations of ignorability



(A) Estimates under no confounding.



(B) Bounds under posited confounding.

FIGURE 3. Estimate (black), bounds (red), and confidence bounds (blue) for the ATE by income quartiles. Confounding scenario: $\rho^2 = 1$; $C_Y^2 \approx 0.04$; $C_D^2 \approx 0.03$. Significance level of 5%.

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